

CORD Receipt-to-Structured-Data Pipeline

Progress note — approaches tried, challenges, and results

Objective

Convert unstructured receipt photos (CORD dataset) into structured fields (store name, date, line items, grand total) and export to CSV/DB. Four extraction strategies were implemented and evaluated against CORD's ground-truth annotations.

Approaches tried

Approach	Engine	Result (test split)
OCR-free structured extraction	Donut (fine-tuned on CORD)	Total accuracy 83.9%; item F1 83.4% (test) — best result by a wide margin
OCR + keyword regex	Tesseract	Total accuracy ~0-1%
OCR + keyword regex	docTR	Total accuracy ~0-1%; CER ~13-15% (OCR itself was fine — parsing logic was the bottleneck)
Perspective restoration + largest-number heuristic	OpenCV + Tesseract	Accuracy 6-20%; contour detection succeeded on <45% of photos

Challenges

- File mismatches: 12-54 images per split have no matching ground-truth JSON (or vice versa), so 5-13% of files had to be excluded before any modeling began.
- Image quality: many photos are faded thermal/dot-matrix prints, low-contrast, wrinkled, or shot at an angle with visible shadow bands crossing the text — conditions that materially degrade both classic CV contour detection and OCR character recognition.
- Schema gaps: the released JSON has no date field at all (only store/menu/total categories), so date extraction cannot be evaluated on this dataset regardless of pipeline quality.
- Ambiguous line semantics: many receipts include both TOTAL and SUB_TOTAL, and separately a TOTAL ITEMS (quantity count) line — all three contain the substring "total," which caused repeated false-positive matches in keyword-based parsing until explicitly excluded.
- Currency formatting: amounts use comma-thousands separators with no decimals (e.g. "174,600"), differing from the decimal-currency assumption ("14.00") in generic receipt-OCR reference code, requiring custom parsing.
- Visual clutter: photos include hands, tables, and irregular backgrounds rather than clean flatbed scans, so classic rectangular-contour detection for perspective correction succeeded on well under half of all images — this is a property of this specific photo set, not a general weakness of computer-vision techniques, which remain valuable for deskewing, denoising, and cropping in more controlled capture conditions.

Conclusion

Donut was the best-performing approach on this dataset by a wide margin: 83.9% total accuracy and 83.4% item F1 on the held-out test split, versus roughly 0-20% accuracy for every OCR-plus-parsing pipeline tried. Its advantage is structural, not just a better OCR engine: it reads structured fields directly from image pixels, so it never depends on linearizing a receipt into flat text or matching keywords like "TOTAL," which is exactly where the other approaches broke down. Classic OCR and computer-vision techniques still have real value in this pipeline (e.g. deskewing and cropping upstream of an OCR or layout model), but for end-to-end structured extraction on noisy, real-world receipt photos, the OCR-free, layout-aware model was the clear winner.